

AI

Artificial Intelligence (AI) is a branch of computer science that focuses on enabling machines or software to perform tasks that typically require human intelligence (or mostly human does).

These tasks include reasoning, learning, decision-making, problem-solving, understanding natural language, and perception (like vision and speech).

ML

Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that focuses on developing algorithms that enable computers or machines or agents to learn from data(experience) and improve their performance over time without being explicitly programmed.

WHY IS EXPLAINABILITY IMPORTANT?

The list keeps going on and on...

IBM Watson AI criticised after giving 'unsafe' cancer treatment advice

Wrongfully Accused by an Algorithm

In what may be the first known case of its kind, a faulty facial recognition match led to a Michigan man's arrest for a crime he did not commit.

Why Amazon's Automated Hiring Tool Discriminated Against Women

**Predictive policing algorithms are racist.
They need to be dismantled.**

<https://www.bbc.com/technology/2018/07/180727-ibm-watson-a-criticised-giving-unsafe-cancer-treatment-advice/>

<https://www.msn.com/2018/04/24/technology/facial-recognition-arrest.html>

<https://www.sdu.org/news/womens-rights/why-amazons-automated-hiring-tool-discriminated-against-women/>

<https://www.technologyreview.com/2018/07/17/100296/predictive-policing-algorithms-racist-dismantled-machine-learning-bias-criminal-justice/>

Why Do We Need It?

- Artificial Intelligence are increasingly implemented in our everyday lives to assist humans in making decisions.
- These trivial decisions can vary from a lifestyle choices to more complex decisions such as loan approvals, investments, court decisions and selection of job candidates.
- Many AI algorithms are a Blackbox that is not transparent. This leads to trustability concerns. In order to trust these systems, humans want accountability and explanation.

Why Do We Need It?

- While the machine learning systems deployed in 2008 were mostly within the products of tech-first companies (i.e. Google, YouTube), the false prediction would result in the wrong recommendation to the application user.
- But, when it is being deployed in other industries such as military, healthcare, finance, it would lead to adverse consequences affecting many lives.
- Thus, we create AI systems that explain their decision making.

WHO CARES ABOUT EXPLAINABILITY?

The Public

Companies Grapple With AI's Opaque Decision-Making Process

Uber, Xerox's PARC, Capital One among organizations investigating how AI solves problems.

Opinion: Artificial intelligence

Beware the rise of the black box algorithm

WHO calls for safe and ethical AI for health

16 May 2023 | Departmental update | Reading time: 2 min (527 words)

How to Build Artificial Intelligence We Can Trust

Computer systems need to understand time, space and causality. Right now they don't.

FORCES > INNOVATION > ENTERPRISE TECH

Building Trust In AI: The Case For Transparency

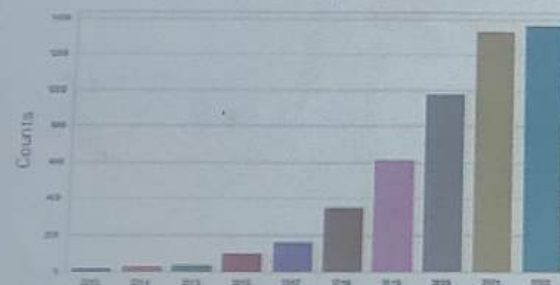
Why businesses need explainable AI—and how to deliver it

September 18, 2020 | Article

WHO CARES ABOUT EXPLAINABILITY?

Academia

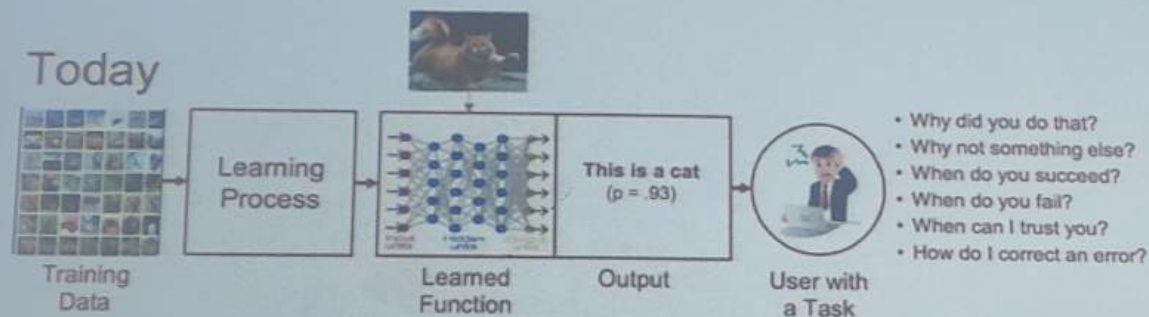
Number of XAI Papers Published



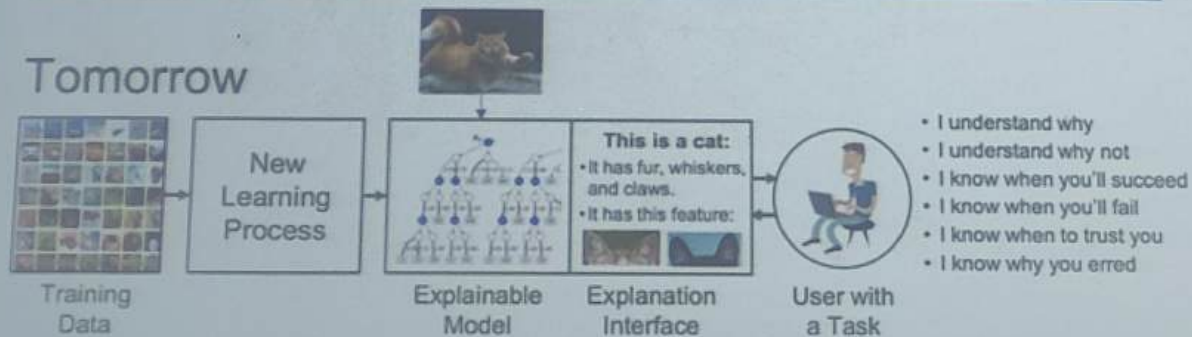
Jacovi, Alon. "Trends in explainable AI (XAI) literature." Medium (2023).

EXPLAINABLE WHAT?

Today



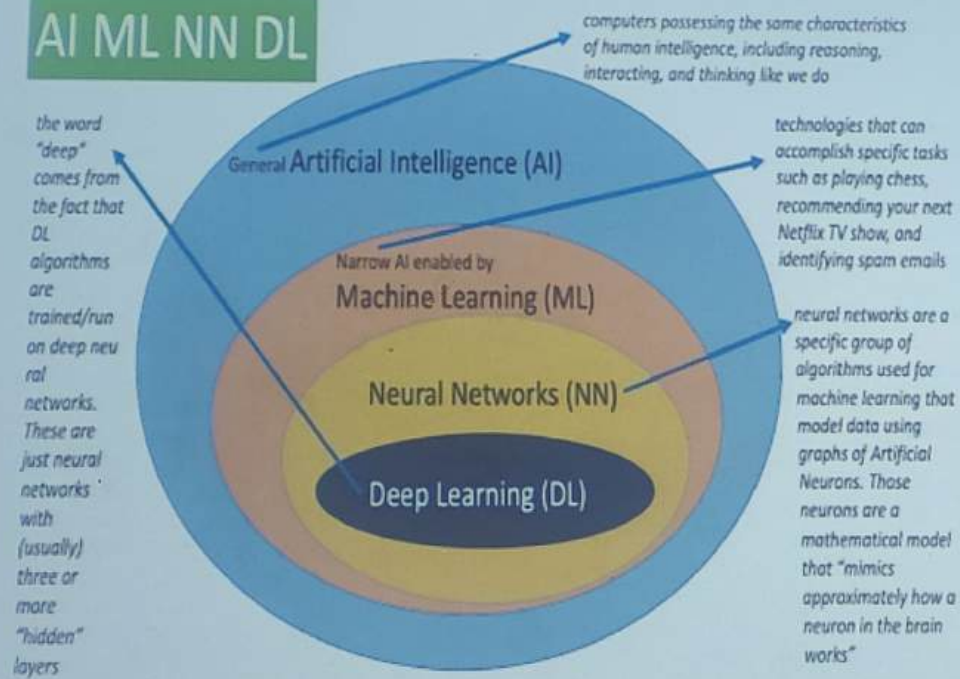
Tomorrow



XAI

Explainable Artificial Intelligence (XAI) is a branch of computer science which refers to a set of techniques and methods that make the output, behavior, and decision-making process of AI systems understandable, justifiable and reliable to humans.

AI ML NN DL



Types of AI

1. Narrow AI (Weak AI)

- Definition: AI that is designed and trained to perform a specific task.
- Behavior: It cannot perform tasks beyond what it is designed for.
- Example:

Siri, Alexa (voice assistants)

Google Translate

Spam email filter

- Note: This is the only type of AI that exists today in real-world applications.

Types of AI

2. General AI (Strong AI)

- Definition: AI with the ability to perform any intellectual task that a human can do.
- Behavior: It would have human-like intelligence and understanding.
- Example:

Theoretical only, not yet achieved.

If built, it could solve math, write poems, drive cars, or diagnose disease without being retrained.

Types of AI

3. Super AI

- Definition: AI that surpasses human intelligence in all respects — creativity, decision-making, emotions, etc.
- Behavior: It would be smarter than the best human minds in every field.
- Example:

Hypothetical/futuristic.

Might control global decisions, innovate faster than humans, or pose ethical risks.

Explainability

Explainability in AI refers to the extent to which the internal mechanics of a machine learning model or AI system can be explained in human terms. It involves making the operations and decisions of AI systems understandable to humans, which is crucial for building trust, ensuring transparency, and fostering accountability.

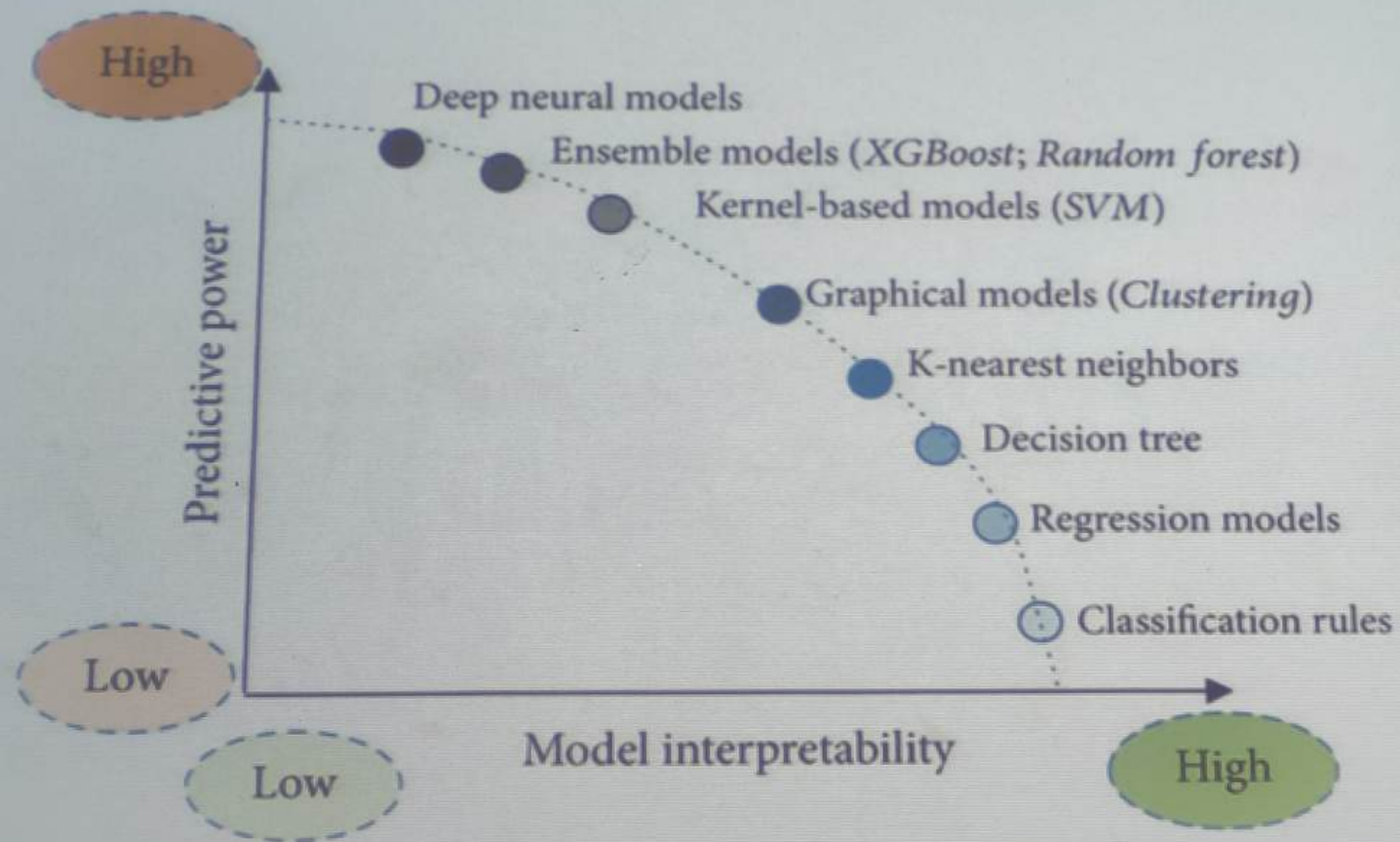
Importance of Explainability

- **Trust and Transparency:** Users are more likely to trust and accept AI systems when they can understand how decisions are made.
- **Regulatory Compliance/Accountability:** Many sectors are subject to regulations that require transparency in automated decision-making processes.
- **Debugging and Improvement:** Explainable models help developers identify and correct errors, biases, and improve the performance of AI systems.
- **Ethical AI:** Ensures AI systems are fair, unbiased, and align with ethical standards, reducing the risk of harm.
- **User Empowerment:** Provides users with insights into AI decisions, enabling them to make informed choices.

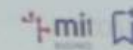
Challenges in Explainability

- **Complexity of Models:** Advanced models like deep neural networks are inherently complex, making them difficult to interpret.
- **Trade-offs with Accuracy:** Simplifying a model for explainability can sometimes reduce its predictive accuracy.
- **Domain-Specific Explanations:** Different stakeholders may require different types of explanations based on their domain knowledge and interests.
- **Dynamic Environments:** AI models that adapt and learn in real-time can change their decision-making processes, complicating explainability.

Trade-off



What is Explainable AI?



Black Box AI



Confusion with today's AI Black Box

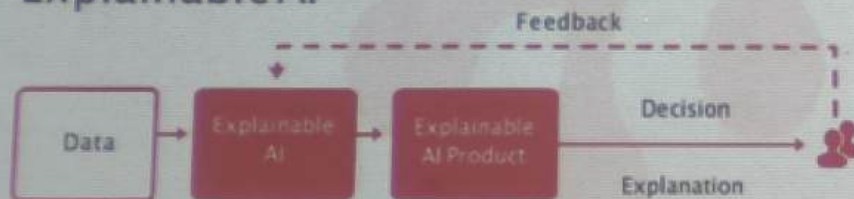
Why did you do that?

Why did you not do that?

When do you succeed or fail?

How do I correct an error?

Explainable AI



Clear & transparent Predictions

I understand why

I understand why not

I know why you succeed or fail

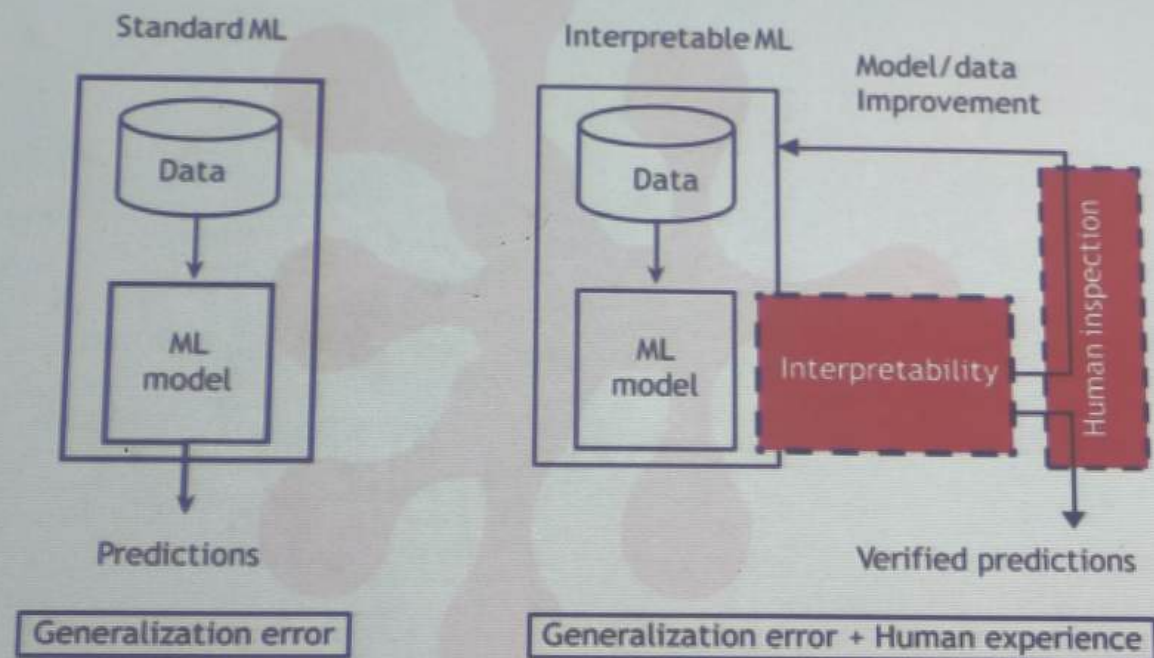
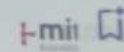
I understand, so I trust you

Objectives/Goals of XAI

(Foundation for Responsible AI)

- **Enhance Transparency**
 - To make the internal logic and behavior of complex AI/ML models understandable to humans.
- **Improve Trust & Acceptance**
 - To increase user confidence in AI systems by providing human-interpretable explanations for predictions or actions.
- **Enable Model Debugging & Improvement**
 - To help developers identify model weaknesses, biases, or errors and refine models more effectively.
- **Support Ethical & Fair Decision Making**
 - To ensure that AI decisions are fair, non-discriminatory, and ethically sound, especially in sensitive domains like healthcare or finance.

Why Explainability: Improve ML Mod



Objectives/Goals of XAI

(Foundation for Responsible AI)

- **Facilitate Compliance with Legal Regulations**
 - To align with data protection laws (like GDPR's "Right to Explanation") by offering justifications for automated decisions.
- **Assist Human Decision-Makers**
 - To help domain experts (doctors, judges, auditors, etc.) make better informed, AI-assisted decisions by understanding model reasoning.
- **Ensure Accountability and Auditability**
 - To provide a basis for explaining decisions during audits, investigations, or legal disputes involving AI.
- **Encourage Responsible AI Deployment**
 - To promote the design of AI systems that are safe, interpretable, and aligned with human values.

Principles of XAI

1. Transparency

AI systems should clearly communicate how they function and why they make certain decisions.

2. Interpretability

The model's predictions and internal workings should be understandable to humans, especially to the intended users.

3. Trustworthiness

Explanations should promote user confidence and reliability.

Models should behave consistently and not manipulate or deceive users.

4. Fairness and Non-discrimination

Explanations should help detect and correct biases or unfair treatment in AI decisions.

Principles of XAI

5. Fidelity (Faithfulness)

Explanations must accurately reflect the model's actual reasoning, not just provide simplified or approximated logic.

6. Robustness

Explanations should remain stable and reliable under small perturbations of input data or model updates.

7. Privacy Preservation

In the process of explaining, the system should not expose sensitive information about users or data used for training.

8. User-Centric Design

Explanations should be tailored to the user's level of expertise, goals, and context (e.g., different for engineers vs end-users).

Limitations of XAI

- Limited interpretability of complex models like deep neural networks
- Trade-off between model accuracy and interpretability
- Not all AI models can be fully explained (e.g., LLMs)
- Explanations may oversimplify or mislead
- Explanations are not always faithful to the actual model logic
- Risk of exposing sensitive or proprietary information

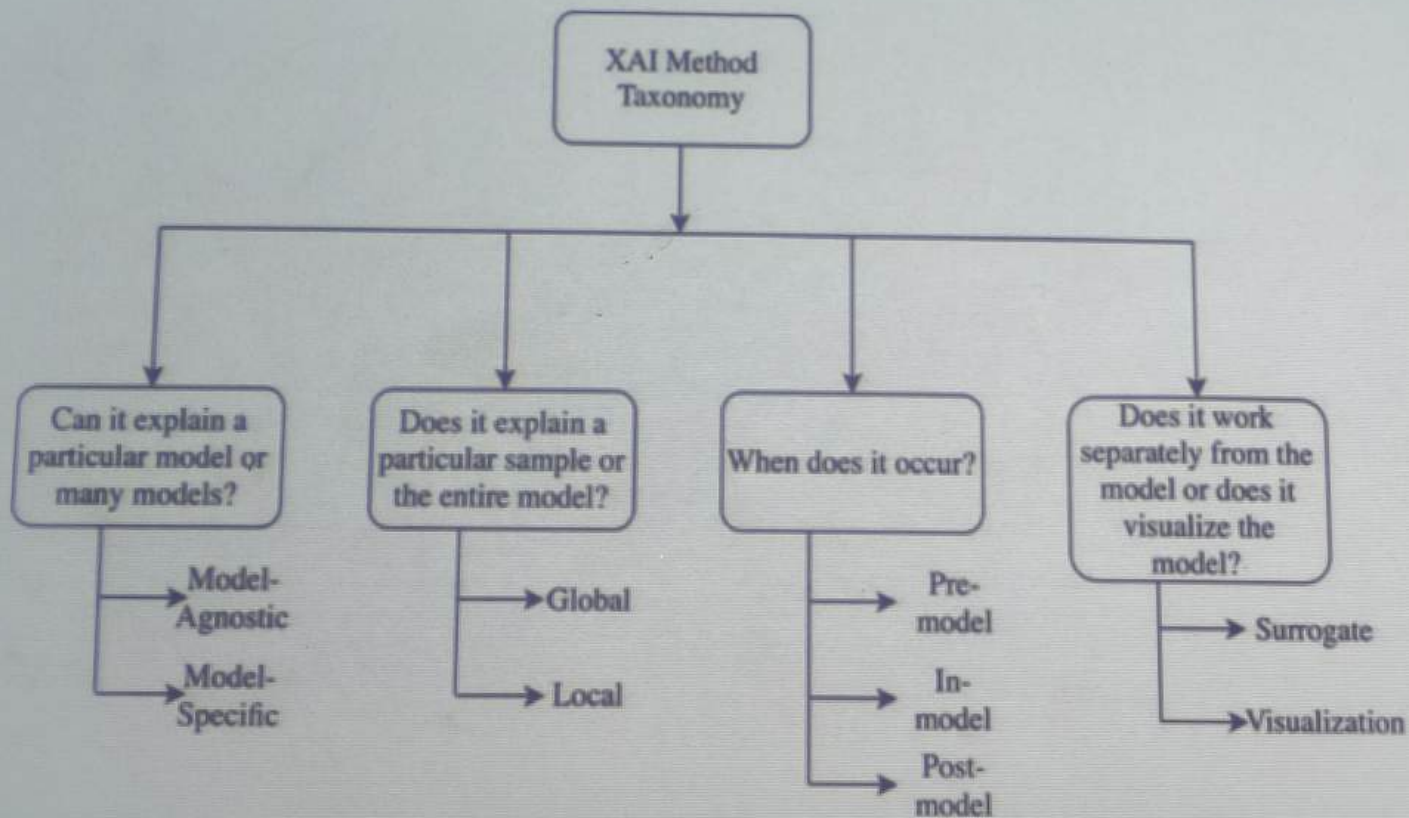
Challenges in XAI

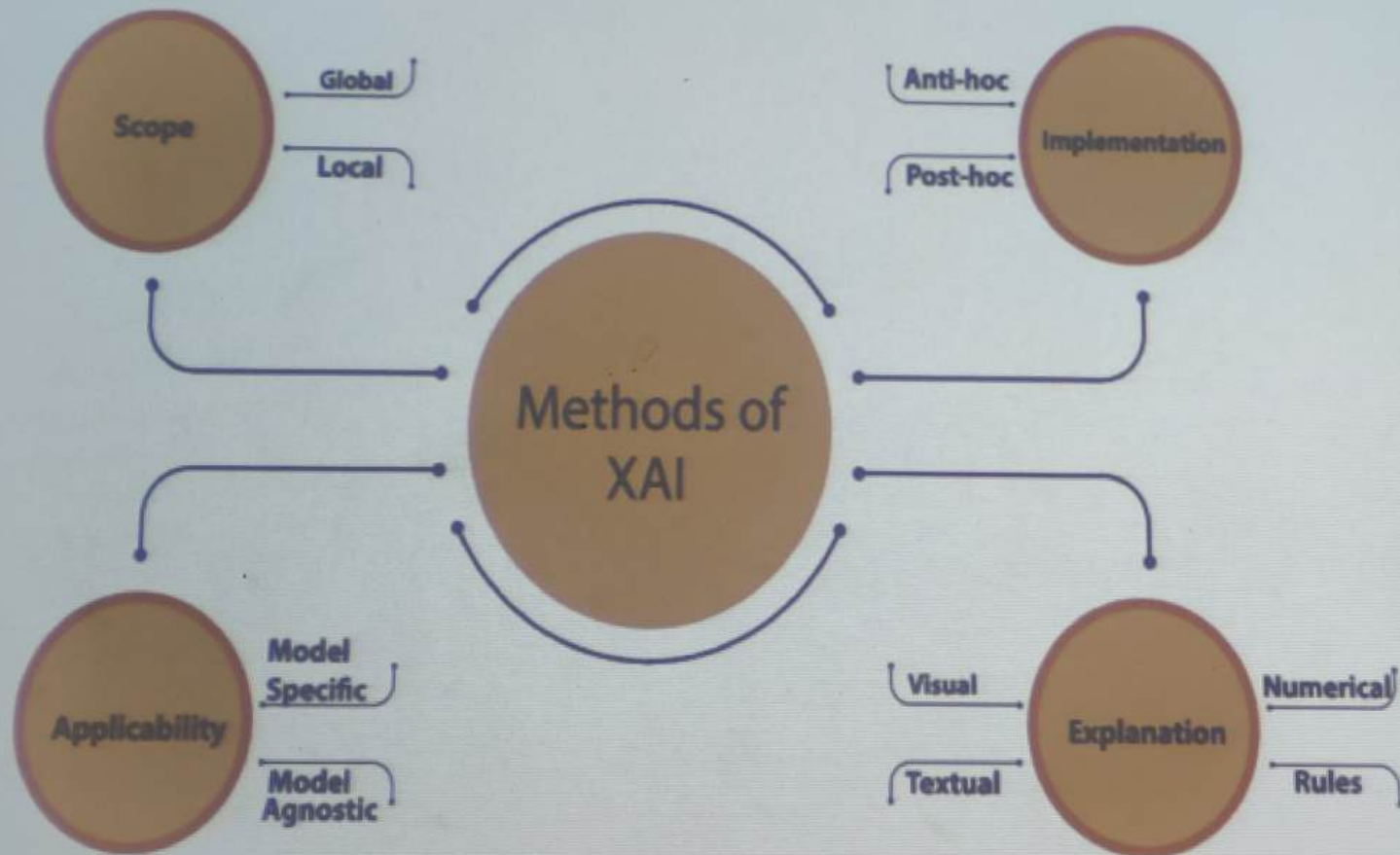
- Designing user-friendly explanations for non-technical users
- Measuring interpretability objectively
- Choosing the right XAI method for the model and context
- Gaining trust while ensuring model accountability
- Adhering to legal and ethical guidelines (e.g., GDPR)

Stakeholders of XAI

Stakeholder	Example	Explanation Need
1. End Users	A loan applicant checking why their application was rejected	Simple, clear explanation: e.g., 'Low credit score and high existing debt'
2. Domain Experts	A doctor using AI to diagnose a disease from MRI images	Medical reasoning: Highlight the features influencing diagnosis
3. AI Developers / Data Scientists	A data scientist debugging a model predicting stock prices	Need to identify model bias, important features, and error patterns
4. Business Decision Makers	A retail manager using AI for inventory prediction	Strategic insight: 'Which product trends are influencing forecast?'
5. Regulatory Bodies / Auditors	A government auditor reviewing an AI-based loan approval system	Legal accountability: Model must explain decisions in an auditable, fair way
6. Legal Professionals / Courts	A judge analyzing AI-assisted parole recommendation	Explanation that is legally sound and understandable for courtroom use

Taxonomy of XAI Techniques





Model-Agnostic:

These methods can be applied to any AI model, regardless of its internal workings. They typically analyze the model's inputs and outputs to understand its behavior. Examples include techniques like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations).

Model-Specific:

These methods are designed for specific types of models and exploit their internal structure for explanation. For example, visualizing activation maps in convolutional neural networks or using decision rules in decision trees.

Local Explanations:

These focus on explaining individual predictions made by the model, providing insights into why a specific input was classified in a certain way.

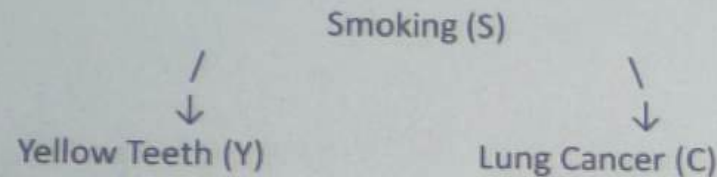
Global Explanations:

These aim to explain the overall behavior of the model, revealing which features are most important for the model's decisions across all inputs.

Causal Model Induction

- Causal model induction is the process of discovering and modeling cause-effect relationships between variables in a dataset, rather than just correlations.
- It's about answering "what causes what?" — not just "what happens together?"
- Causal model often uses structural representation(ex. DAG) to find out underlying mechanism.

Causal Model Induction



- Example:

A predictive model might say:

Smoking and yellow teeth are correlated with lung cancer.

A causal model will say:

Smoking causes both yellow teeth and lung cancer. So, removing "yellow teeth" doesn't affect cancer — but removing smoking might.

Causal Model Induction

Contribution of Causal Induction	Benefit to Explainability
Identifies cause-effect relationships	More meaningful and trustworthy explanations
Enables counterfactual reasoning	Supports "what if" and decision analysis
Reduces reliance on spurious correlations	Prevents misleading explanations
Improves generalization	Robust across new settings or distributions
Enhances fairness and auditability	Highlights potential biases or discriminatory paths